Analog and Digital Electronics

14.08.26

1. Basic electric calculations based on Ohm's Law, unit of voltage, current and resistors.
2. Electrical power calculations
3. Voltage divider
4. Numericals and capacitors
5. DC and AC fundamentals
6. V-I characteristics of diode
7. Rectifiers: half wave and full wave
8. Filtering (capacitor filter basics)
9. Basic of Zener diode, V-I characteristics.
10. Voltage regulator using Zener diode, basic of BJT.
11. Relation between I_C, I_B, I_E (alpha, beta, gamma)
12. Load line of a transistor, Q point.
13. Transistor as a switch
14. Transistor biasing: base bias and VD. Transistor and amplifiers, basic of C-B, C-E, C-C
15. Block diagram of a regulated power supply. Line and load regulation
16. Basic of operational amplifiers, open loop gain, comparators, block diagram of OP-AMP, ideal and practical OP-AMP
17. Basic of inverting and non-inverting amplifiers, negative feedback, virtual ground concept.
18. Basic of passive and active filters.

19.	LPF (low pass filter), HPF (high pass filter), band pass and notch filter
20.	Basic of oscillator
21.	Barkhausen criteria
22.	Phase shift oscillator
23.	Astable and monostable multivibrator using IC 555
24.	Digital electronics basics
25.	Basic gates with truth table
26.	Latch and flip flop
27.	Transistorised series voltage regulator (TSVR)
28.	Three terminal fixed and variable regulators: 78XX, 79XX and LM317